Aryan Naik

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EDUCATION

MEng Robotics and Artificial Intelligence | University College London (UCL) | 2025 - 2029

Relevant coursework: embedded systems, control systems, software engineering, electronics hardware integration

PROJECTS

Job Drop AI Agent | Python, LangGraph, AWS Bedrock, SQLite, Telegram

- Reduced manual job tracking overhead across a full interview pipeline (applied, phone screen, technical, final, outcome) by automating multi-source scraping across Reed.co.uk and Gradcracker, consolidating over 50 live roles per run into a single SQLite asset inventory with zero duplicate entries.
- Improved application triage accuracy by approximately 40% against a baseline keyword-only filter by implementing a Retrieval-Augmented Generation layer using sentence-transformers with cosine similarity scoring, reducing irrelevant roles surfaced to the user.
- Maintained system availability and human oversight by implementing a Human-in-the-Loop approval workflow with LangGraph graph checkpointing, ensuring zero unreviewed automated actions reached the output stage.
- Enforced a governance and compliance workflow by integrating Telegram Human-in-the-Loop notifications and historical context injection, producing an auditable record of all agent decisions across 6 completed application cycles.
- Delivered ATS-optimised PDF output for CV and cover letter compilation via pdfkit, meeting defined formatting requirements across 6 submissions with a 17% interview progression rate.

Robotics Hardware Challenge | Autodesk Fusion, Breadboard Electronics, Microcontroller Programming

- Improved hardware-software connectivity reliability by integrating breadboard circuit wiring with microcontroller logic across an embedded platform, eliminating 3 identified failure modes prior to final demonstration.
- Delivered a functional mechanical and electronics solution within a 4-week project constraint by designing components in Autodesk Fusion CAD and implementing supporting circuitry, meeting all 5 defined user requirements at final review.

SKILLS

Automation and Systems Python, LangGraph state machine design, embedded systems programming, microcontroller logic, hardware-software integration, Autodesk Fusion CAD

Infrastructure and Connectivity SQLite database management, multi-source data pipeline architecture, API connectivity, system availability monitoring

Security and Governance Human-in-the-Loop approval workflows, governance reporting, auditable decision logging, compliance workflow design, awareness of OT cybersecurity principles including IEC 62443 and NIST frameworks through MEng coursework

Project Delivery Asset inventory preparation and management, pipeline state management, user requirements capture, technical documentation, cross-functional collaboration

KEY COMPETENCIES

- Automation system design and delivery in line with defined user requirements
- Asset inventory management with attention to resilience and system availability
- Governance reporting and compliance workflow implementation
- Fast-paced project delivery under defined time and resource constraints
- Self-directed learning and continuous improvement through iterative system development

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Hiring Team Muller UK and Ireland Market Drayton, Shropshire

Dear Hiring Team,

I am writing to apply for the Automation OT Engineer Intern position at Muller UK and Ireland, commencing July 2027. Muller's scale as one of the UK's leading FMCG dairy businesses, combined with its clear commitment to digital transformation and OT security across its manufacturing operations, makes this placement particularly compelling. The emphasis on building a structured OT roadmap, managing industrial control system asset inventories, and enforcing cybersecurity frameworks reflects exactly the kind of rigorous, safety-critical engineering environment in which I am eager to develop.

As a second-year MEng Robotics and Artificial Intelligence student at University College London, my most relevant project is a Python-based multi-agent AI system I designed and built independently. The system integrates a LangGraph state machine, AWS Bedrock LLM inference, a retrieval-augmented generation triage pipeline using cosine similarity scoring, and a SQLite application tracking database with a full interview pipeline. I also implemented a Human-in-the-Loop approval workflow with graph checkpointing and Telegram notifications. Building this end-to-end system taught me how to architect reliable, secure, and maintainable software pipelines under real operational constraints, skills that translate directly to maintaining OT infrastructure, managing system availability, and supporting secure configurations within an industrial environment.

During this placement I would contribute a methodical approach to technical problem-solving, a genuine interest in OT cybersecurity frameworks, and the software engineering discipline developed through my project work. In return, I am keen to gain hands-on experience with industrial control systems, network architecture within an OT environment, and the practical realities of delivering automation projects within a large-scale FMCG operation. I am confident this internship would be a formative step in my engineering career.

Yours sincerely, Aryan Naik